

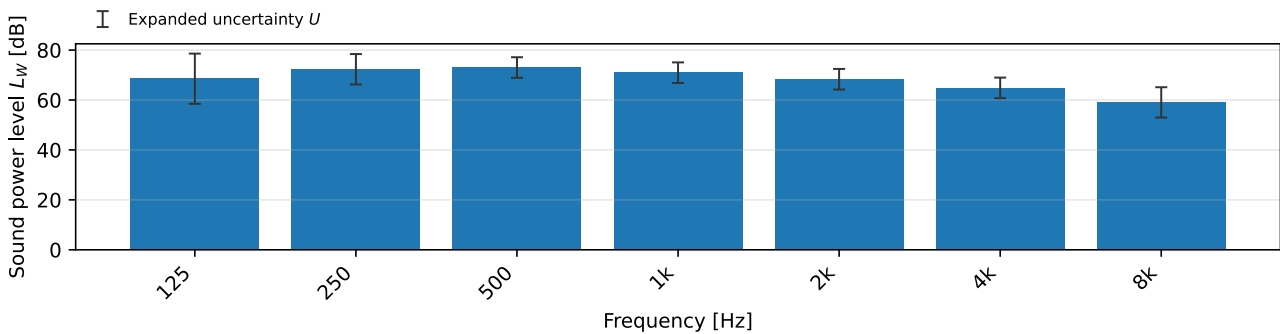
Sound power determination

Determination of the sound power level from the mean sound pressure level in a special reverberation test room, direct method (ISO 3743-2:2018, engineering method, accuracy grade 2).

Client:	Example office equipment maker	Noise source:	Desktop printer, continuous printing
Test environment:	Special reverberation test room, V = 72 m3, Tnom = 0.77 s	Instrumentation:	Class 1 sound level meter (IEC 61672-1), s/n 0042
Temperature [°C]:	22	Relative humidity [%]:	48
Ambient pressure [kPa]:	101.2	Date of test:	2026-09-25

Octave-band sound power levels

f [Hz]	L _p [dB]	L _W [dB]
125	61.8	68.5
250	65.6	72.5
500	66.3	73.0
1000	64.2	71.0
2000	61.6	68.5
4000	58.1	65.0
8000	52.3	59.0



Sound power level $L_{WA} = 75.8 \text{ dB(A) re 1 pW}$

Total $L_W = 78.3 \text{ dB re 1 pW}$
From the octave bands (Annex F): $L_{WA} = 75.8 \text{ dB(A)}$
Determination method: direct method (Formula 9), $V = 72.0 \text{ m}^3$, $T_{nom} = 0.77 \text{ s}$
Expanded uncertainty $U = 4.1 \text{ dB}$, coverage factor $k = 2$

Result vs requirement: $L_{WA} = 75.8 \text{ dB(A)}$, declared limit $\leq 77.0 \text{ dB(A)}$ → **PASS**

The mean level L_p is the energy average over the microphone positions and source locations (Formula 8), each level corrected for background noise by Table 4 at its own position (9.8); $L_W = L_p - 10 \lg(T_{nom}/T_0) + 10 \lg(V/V_0) - 13 \text{ dB}$ (Formula 9). The levels are at the meteorological conditions of the test; $C2 = -0.01 \text{ dB}$ (Annex E) carries them to the reference conditions. The band levels L_W are tabulated to the nearest one-half decibel (12.5 d).

$U = k \sigma_{tot}$; $\sigma_{tot} = \sqrt{(\sigma_{R0}^2 + \sigma_{omc}^2)}$ with $\sigma_{R0} = 2.0 \text{ dB}$ (Table 5, A-weighted) and $\sigma_{omc} = 0.5 \text{ dB}$; $k = 2$ gives a coverage probability of 95 %. The A-weighted sound power level L_{WA} is Formula (9) applied to the mean A-weighted sound pressure level (clause 4); the octave-band total combines the band levels with the corrections of ISO 3743-2:2018 Annex F (Table F.1, Formula F.1). Levels are referenced to the reference sound power 1 pW.

Laboratory: Phonometry reference example

Report no.: EXAMPLE-3743-2

Date: 2026-09-25

Notes: Two source locations for the 500 Hz octave (Table 3).

Operator: phonometry Signature: _____

■ The results relate only to the tested specimen.

Generated by phonometry.